

A Practical Guide to Foundation model AI

TOPIC -- FOUNDATION MODEL AI

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In the United States, the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence defines a foundation model as "an AI model that is trained on broad data; generally uses self-supervision; contains at least tens of billions of parameters; is applicable across a wide range of contexts". In the United States, the proposed AI Foundation Model Transparency Act of 2023 by House Representatives Don Beyer (D–VA) and Anna Eshoo (D, CA) defines a foundation model as "an artificial intelligence model trained on broad data, generally uses self supervision, generally contains at least 1,000,000,000 parameters, is applicable across a wide range of contexts, and exhibits, or could be easily modified to exhibit, high levels of performance at tasks that could pose a serious risk to security, national economic security, national public health or safety, or any combination of those matters." In the European Union, the European Parliament's negotiated position on the E.U. AI Act defines a foundation model as an "AI model that is trained on broad data at scale, is designed for generality of output, and can be adapted to a wide range of distinctive tasks". In the United Kingdom, the Competition and Markets Authority's AI Foundation Models: Initial Report defines foundations model as "a type of AI technology that are trained on vast amounts of data that can be adapted to a wide range of tasks and operations." The United States's definitions are the only ones to make reference to the size of a foundation model, and differ on magnitude. Beyer and Eshoo's definition also specifies that foundation models must achieve a level of performance as to be a potential danger. In contrast, the E.U. definition requires the model to be designed for generality of output. All definitions agree that foundation models must be trained on a broad range of data with potential applications in many domains.

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Systems The size of foundation models also brings about issues with the computer systems they run on. The average foundation model is too large to be run within a single accelerator's memory and the initial training process requires an expensive amount of resources. Such issues are predicted to further exacerbate in future as foundation models grow to new heights. Due to this constraint, researchers have begun looking into compressing model size through tight model inference. GPUs are the most common choice of compute hardware

for machine learning, due to high memory storage and strong power. Typical foundation model training requires many GPUs, all connected in parallel with fast interconnects. Acquiring a sufficient amount of GPUs of requisite compute efficiency is a challenge for many foundation model developers, one that has led to an increasing dilemma in the field. Larger models require greater compute power, but often at the cost of improved compute efficiency. Since training remains time-consuming and expensive, the tradeoff between compute power and compute efficiency has led only a few select companies to afford the production costs for large, state of the art foundation models. Some techniques like compression and distillation can make inference more affordable, but they fail to completely shore up this weakness.

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Modeling For a foundation model to effectively generalize, it must acquire rich representations of the training data. As a result, expressive model architectures that efficiently process large-scale data are often preferred in building foundation models. Currently, the Transformer architecture is the de facto choice for building foundation models across a range of modalities.

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History Quanta Magazine traced world models back to a 1943 publication by Kenneth Craik on mental models and the blocks world of SHRDLU in the 1960s. Business Insider traced world models to a 1971 paper by Jay Wright Forrester. A related idea of organizing world knowledge, the frame representation, was proposed by Marvin Minsky in 1974. In 2018, researchers David Ha and Jürgen Schmidhuber defined world models in the context of reinforcement learning: an agent with a variational autoencoder model V for representing visual observations, a recurrent neural network model M for representing memory, and a linear model C for making decisions. They suggested that agents trained on world models in environments that simulate reality could be applied to real world settings. In 2022, Yann LeCun saw a world model (defined by him as a neural network that acts as a mental model for aspects of the world that are seen as relevant) as part of a larger system of cognitive architecture – other neural networks that are analogous to different regions of the brain. In his view, this framework could lead to commonsense reasoning. LeCun has estimated that world models would be fully functional by the late 2020s to mid 2030s.

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In artificial intelligence, a foundation model (FM), also known as large x model (LxM, where " x " is a variable representing any text, image, sound, etc.), is a

machine learning or deep learning model trained on vast datasets so that it can be applied across a wide range of use cases. Generative AI applications like large language models (LLM) are common examples of foundation models. Building foundation models is often highly resource-intensive, with the most advanced models costing hundreds of millions of dollars to cover the expenses of acquiring, curating, and processing massive datasets, as well as the compute power required for training. These costs stem from the need for sophisticated infrastructure, extended training times, and advanced hardware, such as GPUs. In contrast, adapting an existing foundation model for a specific task or using it directly is far less costly, as it leverages pre-trained capabilities and typically requires only fine-tuning on smaller, task-specific datasets. Early examples of foundation models are language models like OpenAI's GPT series and Google's BERT. Beyond text, foundation models have been developed across a range of modalities—including DALL-E, Stable diffusion, and Flamingo for images, MusicGen and LLark for music, and RT-2 for robotic control. Foundation models are also being developed for fields like astronomy, radiology, ophthalmology, genomics, coding, time series forecasting, mathematics, and chemistry.

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General-purpose AI Due to their adaptability to a wide range of use-cases, foundation models are sometimes considered to be examples of general-purpose AI. In designing the EU AI Act, the European Parliament has stated that a new wave of general-purpose AI technologies shapes the overall AI ecosystem. The fuller structure of the ecosystem, in addition to the properties of specific general-purpose AI systems, influences the design of AI policy and research. General-purpose AI systems also often appear in people's everyday lives through applications and tools like ChatGPT or DALL-E. Government agencies like EU Parliament have identified regulation of general-purpose AI, such as foundation models, to be a high priority. General-purpose AI systems are often characterized by large size, opacity, and potential for emergence, all of which can create unintended harms. Such systems also heavily influence downstream applications, which further exacerbates the need for regulation. In regards to prominent legislation, a number of stakeholders have pushed for the EU AI Act to include restrictions on general-purpose AI systems, all of which would also apply to foundation models.